# A product counterexample to the simplex maximum for projection-body volume

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**Abstract**

The product of two ten-dimensional simplices has larger normalized projection-body volume than a twenty-dimensional simplex. This gives a counterexample to Brannen's proposed simplex maximum.

## 1 Setting and main result

Throughout $d \geq 2$. A *convex body* in $\mathbb{R}^d$ is a compact convex set with nonempty interior, and $|K|$ is its $d$-dimensional volume. We write $S^{d-1}$ for the unit sphere, $B_2^j$ for the Euclidean unit ball of $\mathbb{R}^j$, $\kappa_j = |B_2^j|$, $\mathrm{proj}_{u^\perp}$ for orthogonal projection onto the hyperplane $u^\perp$, and $C + D = \{c + d : c \in C,\ d \in D\}$ for the Minkowski sum. The *support function* of a compact convex set $C$ is $h_C(x) = \max_{y \in C} \langle x, y \rangle$. Two standard facts are used throughout.

(S1) $h_{C+D} = h_C + h_D$; $h_{C \times D}(u, v) = h_C(u) + h_D(v)$; and $h_{MC}(u) = h_C(M^t u)$ for a linear map $M$. All are immediate from the definition.

(S2) $h_C \leq h_D$ implies $C \subset D$; in particular $h_C$ determines $C$. Reason: if $x \notin D$ and $y \in D$ is nearest to $x$, minimizing the distance to $x$ along $[y, z] \subset D$ gives $\langle x - y, z - y \rangle \leq 0$ for all $z \in D$. With $p = x - y$ this says $h_D(p) = \langle p, y \rangle < \langle p, x \rangle$, so $x \notin C$.

The *projection body* $\Pi K = \Pi_d K$ is the compact convex set with

$$h_{\Pi K}(u) = \mathrm{vol}_{d-1}(\mathrm{proj}_{u^\perp} K), \qquad u \in S^{d-1},$$

extended 1-homogeneously to $\mathbb{R}^d$. That such a set exists is classical; (4) below exhibits it for polytopes, and Lemma 4, Part B, re-proves existence for general bodies. The functional studied is

$$R_d(K) = \frac{|\Pi K|}{|K|^{d-1}}, \qquad c_d = \frac{(d+1)d^d}{d!}. \tag{1}$$

Every $d$-simplex has $R_d = c_d$ (Proposition 5). Brannen conjectured in 1996 that simplices maximize $R_d$ among all convex bodies [1], that is,

$$|\Pi K| \leq c_d |K|^{d-1}. \tag{2}$$

**Theorem 1.** *Let $T_{10} = \mathrm{conv}(0, e_1, \ldots, e_{10}) \subset \mathbb{R}^{10}$, where the $e_i$ are the coordinate unit vectors, and let $K = T_{10} \times T_{10} \subset \mathbb{R}^{20}$. Then*

$$\frac{R_{20}(K)}{c_{20}} = \frac{121\binom{20}{10}}{21 \cdot 2^{20}} = \frac{22\,355\,476}{22\,020\,096} > 1.$$

*Thus $K$ violates (2) in dimension twenty.*

**Background.** Petty established the affine covariance of the projection-body operator [6]; see also [5, Equation (1)]. The bound (2) concerns the volume of $\Pi K$ itself, not of its polar; for $d \geq 3$, Petty's separate minimum problem asks whether ellipsoids are the only minimizers of $R_d$; see [7, Section 1]. In dimension three, Chen, Feng, Li, Xi, and Xu settle the minimum problem and prove $R_3(K) \leq 18 = c_3$, with equality for tetrahedra [2, Theorems 1.1 and 1.2]. Feng, Hu, Liu, and Xu give counterexamples to (2) in every dimension $d \geq 9$, using polytopes with at most $d+2$ facets [3, Theorem 1.3], built from perturbed projections of a cube and Minkowski's existence theorem [3, Section 4]. So the failure of (2) was already known; the present paper supplies a direct Cartesian-product witness with its exact value. The optimal upper constant and its maximizers remain separate questions. For centrally symmetric bodies, Henk gives the lower bound $2^d(9/8)^{\lfloor d/3 \rfloor}$ for the optimal upper constant [4]; Corollary 6 below gives an exponential excess over the simplex benchmark $c_n$.

**Proof at a glance.** The key idea is that $R$ is multiplicative under Cartesian products of polytopes, $R_{r+s}(A \times B) = R_r(A)R_s(B)$, while the simplex value is not: $c_{10}^2 > c_{20}$.

- Lemma 2 (facet formula): $\Pi P$ is a sum of segments, one per facet of the polytope $P$. It feeds everything below.
- Proposition 3 (product identity): the facets of $A \times B$ come in two families, so Lemma 2 splits.
- Lemma 4 (affine invariance of $R_d$): used only for the last sentence of Proposition 5, and there only for polytopes (Part A). Part B, for general bodies, is not used by Theorem 1 or Corollary 6.
- Proposition 5 (simplex value): by Lemma 2, $\Pi T_d$ is a scaled cube plus one segment; Fubini gives its volume.
- Theorem 1: Propositions 3 and 5, then integer arithmetic.
- Corollary 6: Proposition 3 iterated, Proposition 5, and $e < 11/4$.

**Renamed symbols.** To avoid clashes, in Lemma 4 the source's radius $R$, factor $\lambda_m$ and functions $F, f_m$ are here $\rho$, $\mu_m$, $\varphi, \varphi_m$; in Corollary 6 the source's $A$ and $d_r$ are $\alpha$ and $q$.

## 2 Facet vectors and Cartesian products

For a facet $F$ of a full-dimensional polytope, $s_F$ is its $(d-1)$-volume, $\nu_F$ its outward unit normal, and $s_F\nu_F$ its *area-normal vector*.

**Lemma 2.** *Let $P \subset \mathbb{R}^d$, $d \geq 2$, be a full-dimensional convex polytope. Then*

$$h_{\Pi P}(u) = \frac{1}{2} \sum_F s_F |\langle \nu_F, u \rangle|, \qquad u \in \mathbb{R}^d, \tag{3}$$

*and therefore*

$$\Pi P = \sum_F \left[ -\frac{s_F \nu_F}{2}, \frac{s_F \nu_F}{2} \right]. \tag{4}$$

*Proof.* Plan: for a unit vector $u$, almost every point of $\mathrm{proj}_{u^\perp} P$ is covered by the projections of exactly two facets; integrating this count gives the factor $\frac{1}{2}$.

*Step 1 (Jacobian).* Fix $u \in S^{d-1}$ and a facet $F$, and let $v_1, \dots, v_{d-1}$ be an orthonormal basis of $\nu_F^\perp$. The projection restricted to the hyperplane of $F$ is affine and multiplies $(d-1)$-volume by the volume $J$ of the parallelepiped spanned by $v_i' = v_i - \langle v_i, u\rangle u \in u^\perp$. Since $u$ is a unit vector orthogonal to all $v_i'$,

$$J = |\det(v_1', \dots, v_{d-1}', u)| = |\det(v_1, \dots, v_{d-1}, u)| = |\langle \nu_F, u\rangle|.$$

The second equality is a column operation. The third expands $u$ in the orthonormal basis $v_1, \dots, v_{d-1}, \nu_F$; only the $\nu_F$-component survives. Hence $\mathrm{vol}_{d-1}(\mathrm{proj}_{u^\perp} F) = s_F|\langle \nu_F, u\rangle|$.

*Step 2 (multiplicity two).* Let $N \subset u^\perp$ be the union of the relative boundary of $\mathrm{proj}_{u^\perp} P$, the projections of all faces of dimension at most $d-2$, and the projections of facets with $\langle \nu_F, u\rangle = 0$. $N$ has $(d-1)$-measure zero: the first set is the boundary of a convex set, and the others lie in finitely many affine subspaces of dimension at most $d-2$. Let $y \in \mathrm{proj}_{u^\perp} P \setminus N$. The line $y + \mathbb{R}u$ meets the interior of $P$; otherwise a hyperplane containing the line would support $P$, and its projection would support $\mathrm{proj}_{u^\perp} P$ at the interior point $y$. So the line meets $P$ in a segment $[p, q]$, $p \neq q$, with $p, q \in \partial P$ and the open segment interior to $P$. Each endpoint lies on a facet, and, since $y$ avoids projections of lower-dimensional faces, on exactly one. These two facets differ, because a facet containing $p$ and $q$ would contain $[p, q]$, which meets the interior. Hence $y \in \mathrm{proj}_{u^\perp} F$ for exactly two facets $F$.

*Step 3 (integrate).* Integrating $\sum_F \mathbf{1}_{\mathrm{proj}\, F} = 2 \cdot \mathbf{1}_{\mathrm{proj}\, P}$ (valid off $N$) over $u^\perp$ and using Step 1,

$$\sum_F s_F|\langle \nu_F, u\rangle| = 2\,\mathrm{vol}_{d-1}(\mathrm{proj}_{u^\perp} P) = 2h_{\Pi P}(u).$$

This is (3) for unit $u$; both sides are 1-homogeneous, so it holds for all $u$.

*Step 4.* The support function of $[-v/2, v/2]$ is $\frac{1}{2}|\langle v, u\rangle|$. By (S1) the right side of (3) is the support function of the right side of (4), and (S2) gives (4). □

**Proposition 3.** *Let $A \subset \mathbb{R}^r$ and $B \subset \mathbb{R}^s$ be full-dimensional convex polytopes, where $r, s \geq 2$. In the orthogonal product space $\mathbb{R}^{r+s} = \mathbb{R}^r \times \mathbb{R}^s$,*

$$\Pi(A \times B) = (|B|\Pi A) \times (|A|\Pi B). \tag{5}$$

*In particular,*

$$R_{r+s}(A \times B) = R_r(A) R_s(B). \tag{6}$$

*Proof. Step 1 (facets).* For $u \in \mathbb{R}^r$ let $A_u = \{x \in A : \langle u, x\rangle = h_A(u)\}$ be the face exposed by $u$; so $A_0 = A$ and $\dim A_u \leq r-1$ if $u \neq 0$. Define $B_v$ likewise. Since $\langle (u, v), (x, y)\rangle = \langle u, x\rangle + \langle v, y\rangle$, the face of $A \times B$ exposed by $(u, v)$ is $A_u \times B_v$, of dimension $\dim A_u + \dim B_v$. A facet is exposed by its outward normal $(u, v) \neq 0$ and has dimension $r + s - 1$, which forces exactly one of $u, v$ to vanish. So the facets are $F \times B$, with unit normal $(\nu_F, 0)$, and $A \times G$, with unit normal $(0, \nu_G)$, where $F$, $G$ run over the facets of $A$, $B$.

*Step 2 (areas).* By Fubini in the orthogonal splitting, $\mathrm{vol}_{r+s-1}(F \times B) = s_F|B|$ and $\mathrm{vol}_{r+s-1}(A \times G) = |A|s_G$. The area-normal vectors are thus $(|B|s_F\nu_F, 0)$ and $(0, |A|s_G\nu_G)$.

*Step 3.* Lemma 2, applied to $A \times B$ and then to $A$ and $B$, gives

$$h_{\Pi(A\times B)}(u, v) = \tfrac{1}{2}\sum_F |B|s_F|\langle \nu_F, u\rangle| + \tfrac{1}{2}\sum_G |A|s_G|\langle \nu_G, v\rangle| = |B|h_{\Pi A}(u) + |A|h_{\Pi B}(v).$$

By (S1) this is the support function of the right side of (5); (S2) proves (5).

*Step 4.* Since $\Pi A \subset \mathbb{R}^r$ and $\Pi B \subset \mathbb{R}^s$, (5) gives $|\Pi(A \times B)| = |B|^r |\Pi A| \cdot |A|^s |\Pi B|$, and $|A \times B| = |A||B|$. Hence

$$R_{r+s}(A \times B) = \frac{|B|^r |A|^s |\Pi A||\Pi B|}{(|A||B|)^{r+s-1}} = \frac{|\Pi A|}{|A|^{r-1}} \cdot \frac{|\Pi B|}{|B|^{s-1}}. \qquad \square$$

The next lemma is used once: to show that all $d$-simplices share one value of $R_d$.

**Lemma 4.** *Let $P \subset \mathbb{R}^d$, $d \geq 2$, be a convex body. For every invertible linear map $L\colon \mathbb{R}^d \to \mathbb{R}^d$,*

$$\Pi(LP) = |\det L| L^{-t} \Pi P, \qquad R_d(LP) = R_d(P),$$

*where $L^{-t}$ denotes the inverse transpose. Translations also preserve $R_d$.*

*Proof.* **Part A: polytopes.** Fix a facet $F$, and write $\nu = \nu_F$, $M = L^{-t}$.

*A1 (normal).* For $x \in P$, $\langle M\nu, Lx\rangle = \langle \nu, x\rangle \leq h_P(\nu)$, with equality exactly on $F$. So $LF$ is a facet of $LP$ with outward unit normal $M\nu/\|M\nu\|$, whatever the sign of $\det L$.

*A2 (area).* The prism $Z = F + [0, \nu]$ has volume $s_F$, so $LZ = LF + [0, L\nu]$ has volume $|\det L| s_F$. $LZ$ is a prism over $LF$ whose height is the component of $L\nu$ along the unit normal from A1:

$$\frac{\langle L\nu, M\nu\rangle}{\|M\nu\|} = \frac{\langle \nu, \nu\rangle}{\|M\nu\|} = \frac{1}{\|M\nu\|}.$$

Volume equals base times height, so $s_{LF} = |\det L| \, \|M\nu\| \, s_F$.

*A3.* By A1 and A2 the area-normal vector of $LF$ is $|\det L| M(s_F \nu_F)$. Since a linear map sends a Minkowski sum of segments to the sum of the image segments, (4) gives $\Pi(LP) = |\det L| M \, \Pi P$.

*A4.* The map $|\det L| M$ has absolute determinant $|\det L|^d |\det L|^{-1} = |\det L|^{d-1}$, so $|\Pi(LP)| = |\det L|^{d-1} |\Pi P|$, while $|LP|^{d-1} = |\det L|^{d-1} |P|^{d-1}$. Hence $R_d(LP) = R_d(P)$.

*A5.* Translating a body changes neither its volume nor its projection volumes; this holds for all convex bodies.

**Part B: general convex bodies** (not needed for Theorem 1 or Corollary 6). Plan: approximate $P$ from inside by polytopes $P_m$ and pass to the limit in Part A.

*B0 (approximants).* By A5 assume 0 is interior to $P$. Choose $a, \rho > 0$ and a cube $P_0$ with $aB_2^d \subset P_0 \subset P \subset \rho B_2^d$. Let $\varepsilon_m \to 0$, let $N_m \subset P$ be a finite $\varepsilon_m$-net (compactness), and put $P_m = \operatorname{conv}(P_0 \cup N_1 \cup \cdots \cup N_m)$. These polytopes increase, contain $aB_2^d$, and with $\mu_m = 1 + \varepsilon_m/a$,

$$P_m \subset P \subset P_m + \varepsilon_m B_2^d \subset \mu_m P_m.$$

The middle inclusion holds because every point of $P$ is within $\varepsilon_m$ of $N_m \subset P_m$. The last holds because $\varepsilon_m B_2^d \subset (\varepsilon_m/a) P_m$ and $P_m + tP_m = (1+t)P_m$ by convexity.

*B1.* For unit $u$ put $\varphi(u) = \operatorname{vol}_{d-1}(\operatorname{proj}_{u^\perp} P)$ and $\varphi_m(u) = h_{\Pi P_m}(u)$. Projecting the sandwich gives $\varphi_m \leq \varphi \leq \mu_m^{d-1} \varphi_m$, and $\varphi_m \leq \kappa_{d-1} \rho^{d-1}$ because $P_m \subset \rho B_2^d$. Therefore

$$0 \leq \varphi - \varphi_m \leq (\mu_m^{d-1} - 1) \kappa_{d-1} \rho^{d-1} \longrightarrow 0.$$

*B2 (existence of $\Pi P$).* The $\varphi_m$ increase, so by (S2) the bodies $\Pi P_m$ increase. They lie in $\kappa_{d-1}\rho^{d-1} B_2^d$ and contain $\kappa_{d-1} a^{d-1} B_2^d$ (as $\varphi_m \geq \kappa_{d-1} a^{d-1}$). The closure of their union is therefore a convex body, with support function $\sup_m \varphi_m = \varphi$; it is $\Pi P$.

*B3.* By B1 and (S2), $\Pi P_m \subset \Pi P \subset \mu_m^{d-1} \Pi P_m$, hence

$$|\Pi P_m| \leq |\Pi P| \leq \mu_m^{d(d-1)} |\Pi P_m|, \qquad |P_m| \leq |P| \leq \mu_m^d |P_m|.$$

*B4 (limit).* Applying $L$ to B0 gives $LP_m \subset LP \subset \mu_m LP_m$, with $LP_m$ increasing polytopes between two fixed balls; so B1–B3 hold for $LP$ and $LP_m$. For every $u$, by B1 for $LP$, Part A with (S1), and B1 for $P$ (extended by homogeneity),

$$h_{\Pi(LP)}(u) = \lim_m h_{\Pi(LP_m)}(u) = \lim_m |\det L|\, h_{\Pi P_m}(L^{-1}u) = |\det L|\, h_{\Pi P}(L^{-1}u).$$

By (S1) the right side is the support function of $|\det L| L^{-t}\Pi P$, so (S2) gives covariance. A4 then gives $R_d(LP) = R_d(P)$; equivalently, B3 for $P$ and for $LP$ lets one pass to the limit in $R_d(LP_m) = R_d(P_m)$. □

## 3 The simplex value

**Proposition 5.** *For $d \geq 2$, the standard simplex $T_d = \mathrm{conv}(0, e_1, \ldots, e_d) \subset \mathbb{R}^d$ satisfies*

$$|T_d| = \frac{1}{d!}, \qquad |\Pi T_d| = \frac{d+1}{((d-1)!)^d}, \qquad R_d(T_d) = c_d.$$

*Every full-dimensional $d$-simplex has the same value $c_d$.*

*Proof.* Plan: list the $d+1$ area-normal vectors, recognise $\Pi T_d$ as a scaled cube plus one segment, and compute that volume by Fubini. Put $w = e_1 + \cdots + e_d$ and $\gamma = 1/(d-1)!$.

*Step 1 (volume).* $T_d$ is a cone of height 1 over $T_{d-1}$, so $|T_d| = |T_{d-1}|/d$; induction gives $1/d!$.

*Step 2 (facets).* $T_d = \{x : x_i \geq 0,\ \sum x_i \leq 1\}$ has $d+1$ facets. The facet in $\{x_i = 0\}$ is a copy of $T_{d-1}$ with outward normal $-e_i$, so its area-normal vector is $-\gamma e_i$. The remaining facet $H \subset \{\sum x_i = 1\}$ has unit normal $w/\sqrt{d}$. Dropping the last coordinate maps $H$ bijectively onto $T_{d-1}$ and, by Step 1 of Lemma 2 with $u = e_d$, multiplies area by $|\langle w/\sqrt{d}, e_d\rangle| = 1/\sqrt{d}$. So $s_H/\sqrt{d} = \gamma$, that is $s_H = \gamma\sqrt{d}$, and the area-normal vector of $H$ is $\gamma w$. (Check: the $d+1$ vectors sum to zero.)

*Step 3 (shape).* By (4), and since a centred segment does not depend on the sign of its generator,

$$\Pi T_d = \sum_{i=1}^{d} \Big[-\frac{\gamma e_i}{2}, \frac{\gamma e_i}{2}\Big] + \Big[-\frac{\gamma w}{2}, \frac{\gamma w}{2}\Big].$$

As $[-v/2, v/2] = [0, v] - v/2$, this is a translate of $\gamma(Q + [0, w])$, where $Q = [0,1]^d = \sum_i [0, e_i]$.

*Step 4 ($|Q + [0,w]| = d+1$).* Let $\hat{w} = w/\|w\|$. For $y \in \mathrm{proj}_{w^\perp} Q$, the fibre $Q \cap (y + \mathbb{R}\hat{w})$ is a segment $I_y$ of some length $\ell(y) \geq 0$, and the fibre of $Q + [0,w]$ over $y$ is $I_y + [0,w]$, of length $\ell(y) + \|w\|$. Over other $y$ both fibres are empty. By Fubini and homogeneity of $h_{\Pi Q}$,

$$|Q + [0,w]| = \int_{\mathrm{proj}_{w^\perp} Q} (\ell(y) + \|w\|)\, dy = |Q| + \|w\|\, h_{\Pi Q}(\hat{w}) = 1 + h_{\Pi Q}(w).$$

The cube has $2d$ facets, of area 1, with normals $\pm e_i$, and $|\langle \pm e_i, w\rangle| = 1$. So (3) gives $h_{\Pi Q}(w) = \frac{1}{2} \cdot 2d = d$.

*Step 5 (value).* Steps 3 and 4 give $|\Pi T_d| = \gamma^d(d+1) = (d+1)/((d-1)!)^d$. Using $(d-1)! = d!/d$,

$$R_d(T_d) = \frac{(d+1)(d!)^{d-1}}{((d-1)!)^d} = \frac{(d+1)(d!)^{d-1} d^d}{(d!)^d} = \frac{(d+1)d^d}{d!} = c_d.$$

*Step 6.* Every full-dimensional $d$-simplex is $LT_d + t$ with $L$ invertible, so Lemma 4 (Part A) gives the last assertion. □

*Sanity check, $d = 2$.* $Q + [0, (1,1)]$ is the hexagon with vertices $(0,0), (1,0), (2,1), (2,2), (1,2), (0,1)$: the unit square plus two parallelograms of area one, total area $3 = d+1$. Then $|\Pi T_2| = 3$ and $R_2(T_2) = 3/(1/2) = 6 = c_2$.

## 4 The counterexample and exponential excess

*Proof of Theorem 1.* $K = T_{10} \times T_{10}$ is a full-dimensional convex polytope in $\mathbb{R}^{20}$. By Proposition 3 and then Proposition 5,

$$R_{20}(K) = R_{10}(T_{10})^2 = c_{10}^2 = \left(\frac{11 \cdot 10^{10}}{10!}\right)^2.$$

Dividing by $c_{20} = 21 \cdot 20^{20}/20!$, and using $10^{20}/20^{20} = 2^{-20}$ and $20!/(10!)^2 = \binom{20}{10} = 184\,756$,

$$\frac{R_{20}(K)}{c_{20}} = \frac{121 \cdot 10^{20} \cdot 20!}{(10!)^2 \cdot 21 \cdot 20^{20}} = \frac{121\binom{20}{10}}{21 \cdot 2^{20}} = \frac{22\,355\,476}{22\,020\,096}.$$

This exceeds 1 because $121 \cdot 184\,756 - 21 \cdot 1\,048\,576 = 335\,380 > 0$. So $|\Pi K| > c_{20}|K|^{19}$, violating (2). □

**Product test.** For $r, s \geq 2$, the same two propositions give $R_{r+s}(T_r \times T_s) = c_r c_s$, and inserting (1),

$$\frac{R_{r+s}(T_r \times T_s)}{c_{r+s}} = \frac{(r+1)(s+1)}{r+s+1}\binom{r+s}{r}\frac{r^r s^s}{(r+s)^{r+s}}.$$

Thus any constant $C$ with $R_{r+s} \leq C$ on all convex bodies in $\mathbb{R}^{r+s}$ satisfies $C \geq c_r c_s$.

**Corollary 6.** *There exist a constant $\lambda > 1$ and an integer $N \geq 20$ such that, for every integer $n \geq N$, some full-dimensional convex polytope $K_n \subset \mathbb{R}^n$ satisfies*

$$\frac{R_n(K_n)}{R_n(T_n)} \geq \lambda^n.$$

*The polytopes $K_n$ may be chosen as Cartesian products of simplices.*

*Proof.* Plan: many factors $T_{10}$ give exponential growth; one further simplex of dimension 10 to 19 reaches every $n \geq 20$; the benchmark $c_n$ is bounded by $(n+1)(11/4)^n$.

*Step 1 (bodies).* Every integer $n \geq 20$ is uniquely $n = 10k + q$ with $k \geq 1$ and $q \in \{10, \ldots, 19\}$. Put $K_n = T_{10}^k \times T_q$, where $T_{10}^k$ is the $k$-fold Cartesian product. Each partial product is a full-dimensional polytope of dimension at least ten, so Proposition 3, applied $k$ times, and Proposition 5 give $R_n(K_n) = c_{10}^k c_q$.

*Step 2 ($e < \beta := 11/4$).* For $j \geq 3$, $1/j! = \frac{1}{6} \cdot 1/(4 \cdot 5 \cdots j) \leq \frac{1}{6}\,3^{-(j-3)}$, strictly for $j \geq 4$. Hence

$$e = \frac{5}{2} + \sum_{j \geq 3} \frac{1}{j!} < \frac{5}{2} + \frac{1}{6}\sum_{i \geq 0} 3^{-i} = \frac{5}{2} + \frac{1}{4} = \beta.$$

*Step 3.* $n^n/n!$ is one term of the positive series $\sum_j n^j/j! = e^n$. So $c_n = (n+1)n^n/n! < (n+1)e^n < (n+1)\beta^n$ for $n \geq 2$.

*Step 4 (exact constants).* Since $10! = 2^8 \cdot 3^4 \cdot 5^2 \cdot 7$,

$$c_{10} = \frac{11 \cdot 2^{10}5^{10}}{10!} = \frac{11 \cdot 2^2 \cdot 5^8}{3^4 \cdot 7} = \frac{17\,187\,500}{567}, \qquad \alpha := \frac{c_{10}}{\beta^{10}} = \frac{2^{22} \cdot 5^8}{567 \cdot 11^9} = \frac{1\,638\,400\,000\,000}{1\,336\,956\,340\,797} > 1.$$

(Here $2^{22}5^8 = 2^{14} \cdot 10^8$ and $11^9 = 2\,357\,947\,691$.)

*Step 5 (comparison).* Fix $\lambda$ with $1 < \lambda < \alpha^{1/10}$. By Proposition 5, $R_n(T_n) = c_n$. Using Steps 1 and 3, then $c_{10}^k = \alpha^k \beta^{10k}$ and $n = 10k + q$,

$$\frac{R_n(K_n)}{R_n(T_n)\lambda^n} > \frac{c_{10}^k c_q}{(n+1)\beta^n \lambda^n} = c_q \beta^{-q} \lambda^{-q} \frac{(\alpha/\lambda^{10})^k}{10k+q+1}.$$

*Step 6 (threshold).* For each of the ten values of $q$, the right side is a positive constant times $\theta^k/(10k+q+1)$ with $\theta = \alpha/\lambda^{10} > 1$, so it tends to infinity with $k$. Choose $k_q \geq 1$ such that it is at least 1 for all $k \geq k_q$, and put $N = \max_q(10k_q + q) \geq 20$. If $n = 10k + q \geq N$, then $k \geq k_q$, and Step 5 gives $R_n(K_n)/R_n(T_n) > \lambda^n$. □

## Verification checklist

1. **The factor $\frac{1}{2}$.** Lemma 2, Steps 2–3: almost every point of the shadow lies under exactly two facets. Every later constant depends on this.
2. **Facets of a product.** Proposition 3, Steps 1–2: only $F \times B$ and $A \times G$ occur, with areas $s_F|B|$ and $|A|s_G$. Then check the exponents in Step 4.
3. **Slanted facet of the simplex.** Proposition 5, Step 2: area-normal vector $w/(d-1)!$.
4. **Cube plus segment.** Proposition 5, Step 4: $|Q + [0, w]| = 1 + h_{\Pi Q}(w) = d + 1$; compare the hexagon for $d = 2$.
5. **Simplex algebra.** Proposition 5, Step 5; test $c_2 = 6$.
6. **The integer inequality.** Proof of Theorem 1: $121 \cdot 184\,756 > 21 \cdot 2^{20}$.
7. **Prism height.** Lemma 4, A1–A2. This is needed only to say that every 20-simplex, not just $T_{20}$, has value $c_{20}$.
8. **Corollary only.** Steps 2 and 4 ($e < 11/4$, $\alpha > 1$) and the cancellation in Step 5.

## References


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